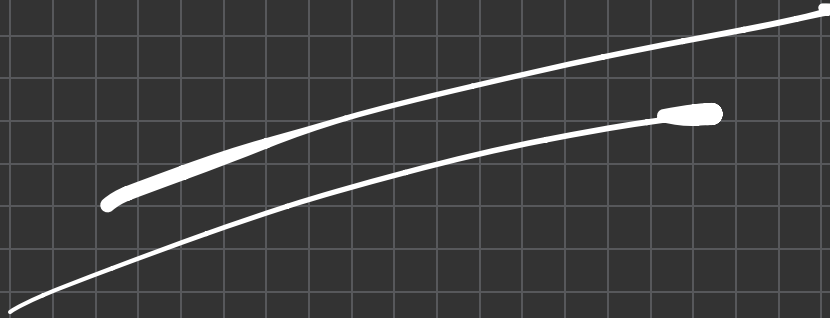
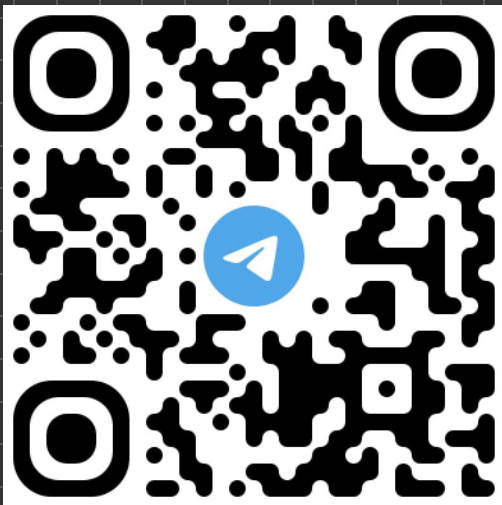


(Lecture 09)

CSS



Telegram



Discord



HTML vs CSS

Let's understand with a real life example

HTML

HTML is like the structure of a car.

It defines the basic parts and arrangement of the car.



Example



Just like HTML creates the basic structure of a webpage using elements like `<html>`, `<head>`, `<body>`, `<div>`, `<p>`, `<h1>` etc.



CSS

CSS is like the style (looks) of a car.

It adds colors, design, shape, and makes the car look attractive.



Example



Just like CSS adds colors, fonts, spacing, backgrounds, borders and makes the webpage look beautiful.

HTML vs CSS



Let's understand with a real life example of a House



THIS IS HTML

HTML is like the **structure (skeleton)** of a house.
It defines the basic parts and arrangement.



In a House

- ✓ Foundation, pillars, walls, roof, doors, windows decide the structure.
- ✓ Everything is in place but it looks plain and incomplete.
- ✓ That's **HTML**.

Example (HTML)

```
<body>
  <h1>Welcome to My House</h1>
  <p>This is my simple house.</p>
</body>
```



THIS IS CSS

CSS is like the **style (looks and design)** of a house.
It adds colors, design, and makes it beautiful.



In a House

- ✓ Paint, furniture, lighting, tiles, decoration make the house beautiful.
- ✓ Same structure, but the look is completely different.
- ✓ That's **CSS**.

Example (CSS)

```
body { background-color: #f5f5f5; font-family: Arial; }
h1 { color: #2c3e50; text-align: center; }
p { color: #555; font-size: 18px; line-height: 1.6; }
```



Summary: HTML builds the **structure**.

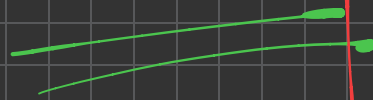
CSS **styles** and beautifies it.



Together, they build a complete house (web page).

CSS

cascading style sheet



formatting
appearance

Types of CSS

3 types CSS

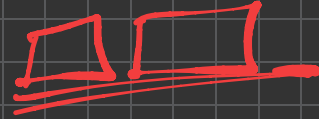
→ Inline CSS

→ Internal CSS

→ External CSS

Inline CSS

Inline



→ elements ke andar he CSS likhte hai

Inline CSS is used to apply a unique style to a single HTML element by using the style attribute directly within the HTML tag.

Inline CSS seedha HTML element ke andar style attribute mein likhte hain

`<p style="" > - - - </p>`

↓
attribute

Color → Red

Internal CSS

↓ ko head tag ke andar
with the help of style tag
likhte hai

Internal CSS (also called embedded CSS) is a method of applying styles to a single HTML page by placing all code inside a `<style>` tag within the `<head>` section of the document.

Internal CSS HTML file ke andar hi hota hai lekin `<head>` mein - `<style>` tag ke andar

External CSS

↓ External CSS file banate hai .html

• CSS → Extension

index.css

External CSS is a web development method where style rules are stored in a completely separate file with a .css extension and linked to an HTML document using the <link> tag.

External CSS mein styles ek alag .css file mein likhi jaati hain
HTML file mein sirf ek <link> tag se us file ko connect karte hain

External

stylesheet

tag

HTML
file

<link>

CSS
file

```
<link rel="stylesheet" href="index.css">
```

<link rel="" href="" >
↓
relation
↓
address

→ Code Repeat ~~X~~

HTML

style

HTML

style

style

index.css

Selectors

CSS selectors are patterns used to target and style specific HTML elements on a web page.

Hum CSS rule likhte hain toh pehla kaam hota hai kisko style karna hai. Woh kaam selector ka hai.

kisko ← selector {
 property : value;
 ↓
 kya
}

kongsi

Element Selector

`<p> — <p>`

`<h1> Hello </h1>`
`p, h1 {`
`}`
`}`

An element selector (also known as a type selector) in CSS selects HTML elements directly by their tag name.

Class selector

new attribute

class = "hello"

• hello
↑

classname

→ class = "nulla"
h5
h6
a } → hilla

↓

5 → p

3 → p

class
hilla

```
• hilla {  
  color : blue  
  
}
```

The CSS class selector targets HTML elements that have a specific class attribute, allowing you to apply identical styling to multiple elements across a webpage. In your CSS file, it is defined by writing a period (.) character followed immediately by the class name.

ID Selector

↓
unique

`<p id="" >` — `<p>`

↓
css ⇒ #sagar

↳ → nilla
↳ → hara
↳ → Pilla

An ID selector in CSS matches and styles a single, unique HTML element based on the value of its id attribute. To target an ID in your stylesheet, you must prefix the ID name with a hash symbol (#).

Universal selector

* {
→ CSS property
}

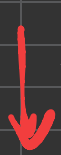
The CSS universal selector (*) matches and selects every single HTML element on a webpage. It acts as a wildcard, applying its specified styles globally to all tags, including <html>, <body>, containers, headings, and paragraphs.

specificity

inline css > id > class > Element > *

CSS specificity is the algorithm browsers use to determine which style rule takes precedence when multiple selectors target the same HTML element. The rule with the highest specificity "wins" and gets applied.

Combinators



Elements ke beech ka relation

<p> , H1 , H2

CSS combinators are symbols or characters used to combine multiple selectors to define a specific relationship between elements in the HTML structure.

Parents

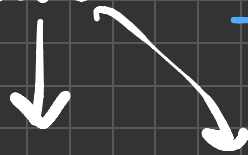


(Tum)

child

child

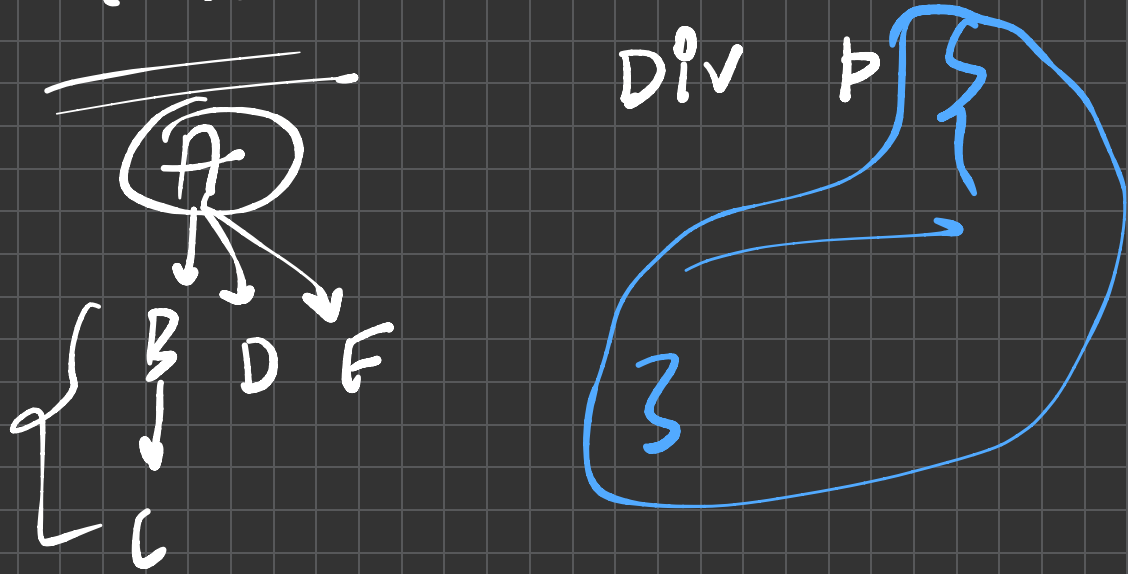
child



child

child

Descendant Combinator



The descendant combinator in CSS is represented by a single space () character. It is used to select and target all elements that are nested inside a specified ancestor element, regardless of how deeply nested they are.

child combinator

Div > p {

3

The CSS child combinator (>) targets only the elements that are direct, immediate children of a specified parent element. It prevents styles from trickling down to deeply nested grandchildren or further descendants.

Adjacent sibling combinator

↓
ek dum next a + b {
}

The adjacent sibling combinator (+) in CSS matches the second element only if it immediately follows the first element, and both elements share the same parent.

general sibling combinator

$a \sim b$

}

The general sibling combinator ($\sim$) in CSS matches all occurrences of a second element that follow a first element and share the same parent element.

Combinator	Symbol	Meaning
Descendant	$A\ B$	B andar hai A ke (kisi bhi depth)
Child	$A > B$	B seedha direct child hai A ka
Adjacent Sibling	$A + B$	B bilkul next sibling hai A ke
General Sibling	$A \sim B$	B baad ke koi bhi sibling hai A ke

Pseudo class :

Element → state based styling

selector : pseudo-class {

A CSS pseudo-class is a keyword added to a selector that defines a special state of the selected element.

}

Pseudo-class ek element ki current state ke basis pe style karta hai

$$2n+1$$

$$2 \times 0$$

$$0+1$$

$$1$$

$$2 \times 1 + 1$$

$$2+1$$

$$3$$

$$a_n + b_n$$

$$a_n$$

$$2 \times 1$$

$$0$$

$$2$$

$$4$$

$$6$$

$$8$$

$$0$$

$$1$$

$$2$$

$$3$$

$$4$$

$$5$$

$$1$$

$$2$$

$$3$$

Pseudo Elements

↓ ke
content ke part
ko style karo

A CSS pseudo-element is a keyword added to a selector that lets you style a specific part of an HTML element.

Pseudo-element ek element ke specific part ko target karta hai

selector :: pseudo-element {

}

	Pseudo-class	Pseudo-element
Syntax	Single colon :	Double colon ::
Targets	Element ki state	Element ka part/ content
Example	:hover, :nth-child	::before, ::after